

Amit Agarwal

Senior Principal Applied Scientist & Technical Lead

Enterprise AI Systems, Agentic AI, Multimodal Models, Post-Training & Evaluation

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Professional Summary

Applied scientist and technical lead with **10+ years** spanning AI research, system architecture, and enterprise delivery. Connects science, engineering, and product teams across support agents, multimodal and vision-language systems, video analytics, document intelligence, and conversational search.

- **Research recognition:** Co-author of *AccessEval* (**EMNLP 2025 Social Impact Award**) and *Judging What We Cannot Solve* (**ICML 2026 Spotlight**).
- **Research & IP:** Research across ACL, EMNLP, ICML, ICLR, CVPR, and a NeurIPS workshop; inventor on **10 issued patents**, with **41 pending**.
- **Technical leadership:** Built and grew applied-science teams from inception through architecture and delivery of enterprise semantic layers, personalized search and agents, and support automation; **lead workshop organizer** at CVPR, ACL, and EMNLP.

Experience

Oracle Cloud Infrastructure (OCI)

Apr 2021 - Present

Senior Principal Applied Scientist (current), Generative AI

Seattle, WA

- Develop support agents for cloud-console assistance, ticket triage, and DevOps incident resolution. Architect Agent Hub routing, skill/tool retrieval, and shared context and memory for long-horizon tasks.
- Developed a personalized leadership-coaching agent pilot combining assessment-driven plans, persistent memory, retrieval-grounded recommendations, role-play practice, and proactive nudges.
- Owned applied-science contributions to multimodal and agentic enablement of **Oracle Fusion ERP** and **Oracle AI Database 26ai**: multimodal RAG, embedding fusion, image understanding, and text-to-image capabilities with product engineering.
- Architected an enterprise knowledge and semantic layer for multi-agent workflows and conversational search, combining knowledge graphs, multimodal grounding, and organization-level data isolation.
- Led multimodal and multilingual embedding and ranking model development across documents, code, images, audio, and video, supported by synthetic-data tooling.
- Lead evaluation research and tooling for retrieval, multimodal reasoning, consistency, and responsible AI; build benchmarks and harnesses to assess safety, bias, and content moderation and guide model and system decisions.
- Led a multi-agent platform spanning RAG, code, and enterprise workflows across **15+ engagements**; established tool-use, customer-feedback, and evaluation loops.
- Delivered OCR, visual question answering, layout understanding, and key information extraction to **20+ customers**; advanced low-data document AI with graph neural networks and synthetic data.

Jio Platforms (Reliance)

Jun 2020 - Apr 2021

Data Science Lead, Computer Vision

Hyderabad, India

- Led video-analytics development from data and model design through evaluation and product integration.
- Designed detection, tracking, scene understanding, and security/compliance for indoor and outdoor cameras.
- Built explainable AI for medical imaging at Reliance Hospitals and OCR-based document understanding and information retrieval for Reliance Retail and Hospitals.

Abzooba Infotech (acquired by UST Global)

Jun 2016 - Jun 2020

Data Scientist

Pune, India

- Served as product owner and technical lead for financial-news recommendation and shipping-disclosure classification systems.
- Developed query understanding, named entity recognition, word-sense disambiguation, and implicit-feedback models for enterprise search and recommendation.
- Built document search, summarization, chart and entity image disambiguation, customer-feedback analytics, and predictive workflows for a Singapore government SmartCity program.

Selected Publications

- **Aligning LLMs for Multilingual Consistency in Enterprise Applications**
EMNLP 2025. First author. Batch-aligned preference optimization; Llama-3.1-70B achieved **74.9% average MGSM exact match vs. 57.2%** for the baseline in the reported study.
- **Do Image-Text Metrics Respect Semantic Invariances?**
ACL 2026. First author. Evaluated five image-text metrics under meaning-preserving changes; post-hoc calibration roughly halved median sensitivity in the reported experiments.
- **RCI: A Score for Evaluating Global and Local Reasoning in Multimodal Benchmarks**
EMNLP 2025. First author. Compared patch-level and full-image performance across **13 multimodal benchmarks** to diagnose reliance on localized cues and spatial biases.
- **Hard Negative Mining for Domain-Specific Retrieval in Enterprise Systems**
ACL 2025. Co-first author (equal contribution). Hard-negative mining for enterprise rerankers; reported **MRR@10 of 0.64 vs. 0.57** for ADORE+STAR on a cloud-services test set.
- **Judging What We Cannot Solve**
ICML 2026. Co-author. Consequence-based, oracle-free evaluation of research-level mathematics, examining how to assess answers when reference solutions are unavailable.
- **AccessEval: Benchmarking Disability Bias in Large Language Models**
EMNLP 2025. Co-author. Evaluated **21 LLMs across 6 domains and 9 disability types**, using paired queries to measure changes in factual accuracy, sentiment, and social perception.

Additional work spans ACL, EMNLP, NAACL, CVPR, ICLR, COLING, DASFAA, and CIKM. Full publication list: [Google Scholar](#).

AI Research & Systems Expertise

Agent systems & architecture: Multi-agent orchestration, agent routing, skill and tool retrieval, context and memory, tool-use and feedback loops, retrieval-augmented generation (RAG).

Model adaptation & post-training: Multimodal and multilingual modeling, vision-language models (VLMs), supervised fine-tuning, direct preference optimization (DPO), RLHF, reward modeling.

Evaluation science: Benchmark and metric design, retrieval, multimodal reasoning and consistency, semantic invariance, context robustness, responsible AI, safety, bias, content moderation.

Research engineering: Reproducible experiments, synthetic-data pipelines, agent evaluation harnesses, reasoning and tool-use evaluation, customer-feedback loops, model and system integration.

Education

MS, Machine Learning & Artificial Intelligence, Distinction, Liverpool John Moores University

BTech, Computer Science & Engineering, IEM, West Bengal University of Technology

Research Service & Open Collaboration

- **Lead workshop organizer:** GRAIL-V at CVPR, SURGeLLM at ACL, and DocInsights at EMNLP.
- **Research service:** Area chair, program committee member, and reviewer across AAAI, NeurIPS, ACL, CVPR, EMNLP, NAACL, IJCAI, and ARR.
- **Mentoring and talks:** Speaker at MIPR 2024 and ICDMAI 2020; Course mentor with MIT and UT Austin programs.
- **Open source:** Contributions and collaborations across VLMEvalKit, EasyOCR, PyTorch Lightning, and Flair.

Additional Honors

Oracle Edison Award | Academic Star, IIIT Bangalore | **Employee of the Year**, Abzooba